

Paper-to-Peer (P2P)

— Matrix-Weighted Consensus — and its applications

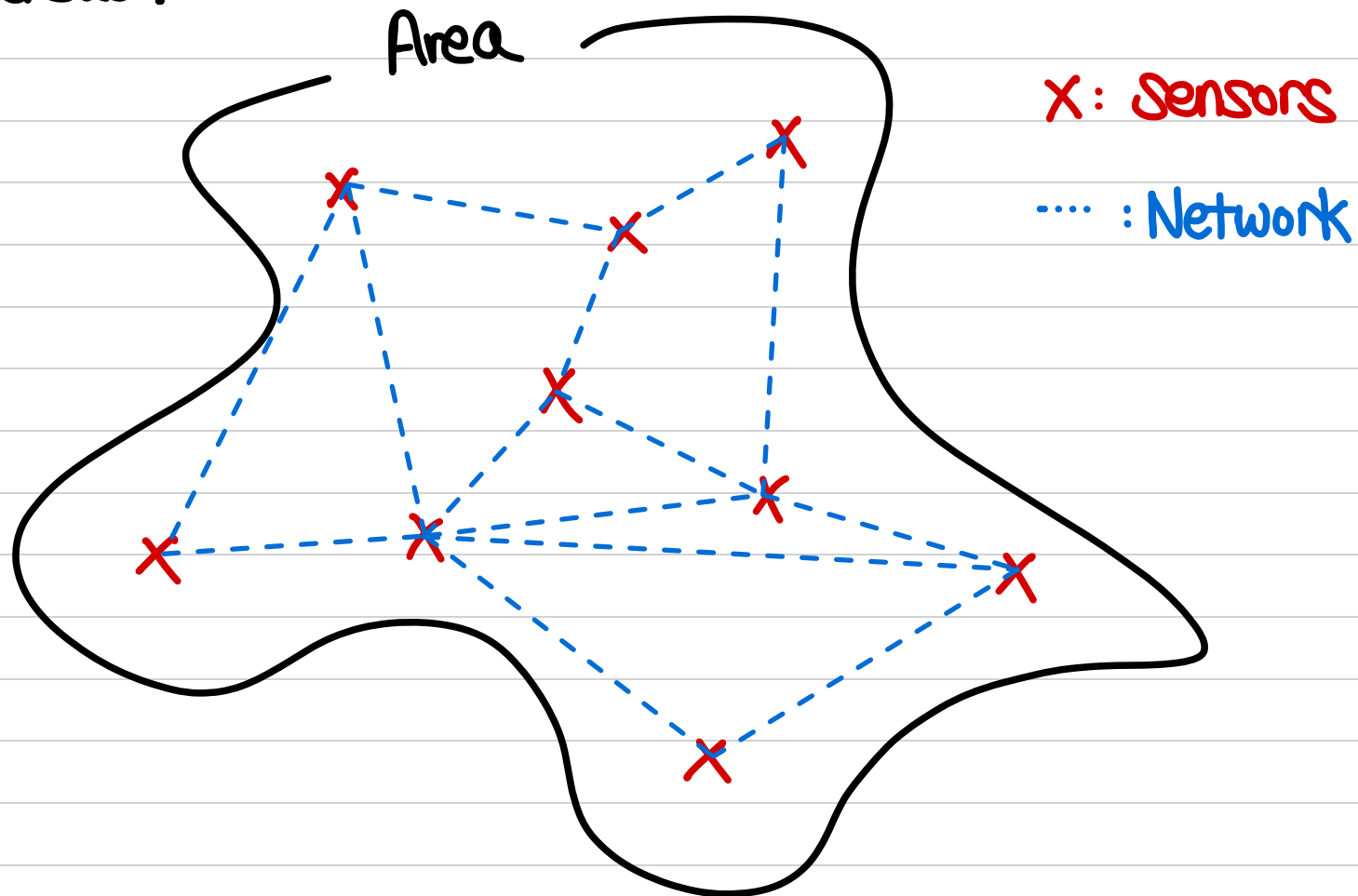
Sep. 14. 2026 (Week 2)

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* Overview

- Consensus?



Q. What is the single representative value of temperature?

Design Consensus Protocol

Assumption:

(The rate of change of each unit's state is governed by)
(the **sum of its relative states w.r.t. neighboring units.**)

Ex

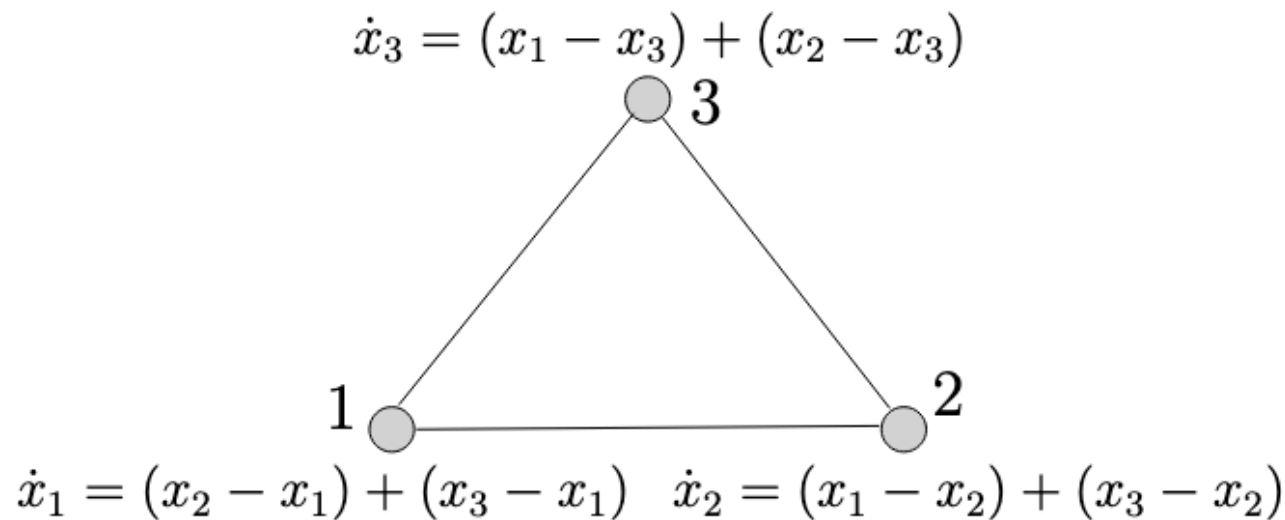


Figure 3.1: Agreement protocol over a triangle

In general, $\dot{x}_i(t) = \sum_{j \in N(i)} (x_j(t) - x_i(t)) \quad \forall i = 1, \dots, n$

→ Matrix form: $\dot{x}(t) = -L(G)x(t)$

$$\text{Why? : } \dot{x}_i(t) = \sum_{j \in N(i)} (x_j(t) - x_i(t))$$

$$= \left(\sum_{j \in N(i)} x_j(t) \right) - \deg(i) x_i(t)$$

$$= \left(\sum_{j=1}^n \underbrace{x_j(t)}_{\substack{\downarrow \\ \text{Off-diagonal}}} \times \underbrace{1(j \in N(i))}_{\substack{\downarrow \\ \text{Adjacency}}} \right) - \underbrace{\deg(i)}_{\substack{\downarrow \\ \text{Degree}}} \underbrace{x_i(t)}_{\substack{\downarrow \\ \text{Diagonal}}}$$

∴ By concatenating, $\dot{x}(t) = (A(G) - \Delta(G))x(t) = -L(G)x(t)$

- This protocol is designed **Manually** (Not from physical law!)

Q. Is this protocol **Natural**?

① Simple local feedback

- ↳ Agents only can access neighbor's info. (**lack of global info.**)
 - Only react w.r.t. **relative state** to local neighbors

② Achieve our objective

- ↳ To reach consensus, $\left\{ \begin{array}{l} \text{my state} > \text{neighbor's state} \rightarrow \text{decrease} \\ \text{my state} < \text{neighbor's state} \rightarrow \text{increase} \end{array} \right.$

③ Physical analogy



Remark: Physical meaning of $L(G)$

• Heat equation: $\frac{\partial u}{\partial t} = \alpha \nabla^2 u$

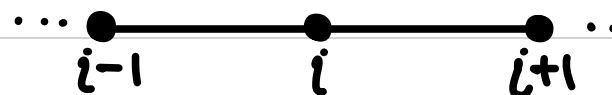
1D rod



Discretize



Graph



↳ 1D ver: $\frac{\partial u}{\partial t} = \alpha \frac{\partial^2 u}{\partial x^2}$

$$\left. \frac{\partial^2 u}{\partial x^2} \right|_i \approx u_{i+1} - 2u_i + u_{i-1}$$

$$= (u_{i+1} - u_i) + (u_{i-1} - u_i)$$

$$\text{So, } \dot{u}_i = \sum_{j \in N(i)} (u_j - u_i)$$

$$\downarrow$$
$$\dot{u} = -L(G)u$$

→ Graph Laplacian is a Discrete Analogue of Laplacian operator. 6/34

Proof) $\dot{x}(t) = -L(G)x(t)$

$$e^{-L(G)t} = e^{-(U\Lambda U^T)t} \text{ (eigendecomposition)}$$

Since $e^A = I + A + \frac{A^2}{2!} + \frac{A^3}{3!} + \dots$, by letting $A = -U(\Lambda t)U^T$,

$$A^n = (-1)^n U(\Lambda t)^n U^T \quad (\because U \in O(n))$$

and hence,

$$\begin{aligned} e^{-(U\Lambda U^T)t} &= I + (-1)U(\Lambda t)U^T + \frac{1}{2!}U(\Lambda t)^2U^T + \frac{1}{3!}(-1)U(\Lambda t)^3U^T + \dots \\ &= U \left(I - (\Lambda t) + \frac{1}{2!}(\Lambda t)^2 - \frac{1}{3!}(\Lambda t)^3 + \dots \right) U^T \\ &= U e^{-\Lambda t} U^T \end{aligned}$$

Here, since $\Lambda = \text{diag}(\lambda_1(G), \lambda_2(G), \dots, \lambda_n(G))$,

$$e^{-\Lambda t} = \text{diag}(e^{-\lambda_1(G)t}, e^{-\lambda_2(G)t}, \dots, e^{-\lambda_n(G)t}).$$

$$\begin{aligned} \text{Thus, } \mathbf{U}e^{-\Lambda t}\mathbf{U}^T &= \begin{pmatrix} | & & | \\ u_1 & \cdots & u_n \\ | & & | \end{pmatrix} \begin{pmatrix} e^{-\lambda_1(G)t} & & 0 \\ & \ddots & \\ 0 & & e^{-\lambda_n(G)t} \end{pmatrix} \begin{pmatrix} -u_1^T - \\ \vdots \\ -u_n^T - \end{pmatrix} \\ &= e^{-\lambda_1(G)t} u_1 u_1^T + e^{-\lambda_2(G)t} u_2 u_2^T \cdots + e^{-\lambda_n(G)t} u_n u_n^T. \end{aligned}$$

By letting $x(0) = x_0$ as an initial condition,

$$\begin{aligned} x(t) &= e^{-\mathbf{A}(G)t} x_0 \\ &= \left(e^{-\lambda_1(G)t} u_1 u_1^T + e^{-\lambda_2(G)t} u_2 u_2^T \cdots + e^{-\lambda_n(G)t} u_n u_n^T \right) x_0 \end{aligned}$$

Recall · $\lambda_1(G) = 0$ and corrsponding eigenvector $u_1 = \frac{1}{\sqrt{n}} \mathbf{1}$ (normalized one vec)

$$\lambda_2(G) > 0$$

$$x(t) = \underbrace{e^{-\lambda_1(G)t}}_{\substack{\parallel \\ 1}} \underbrace{u_1}_{\substack{\parallel \\ \mathbf{1}/\sqrt{n}}} \underbrace{u_1^T}_{\substack{\parallel \\ \mathbf{1}^T/\sqrt{n}}} x_0 + \underbrace{e^{-\lambda_2(G)t} u_2 u_2^T x_0 + \dots + e^{-\lambda_n(G)t} u_n u_n^T x_0}_{\rightarrow 0 \text{ as } t \rightarrow \infty \text{ by exponential.}}$$

$$= \frac{1}{n} \mathbf{1} \underbrace{\mathbf{1}^T x_0}_{\text{Scalar}} = \frac{1}{n} (\mathbf{1}^T x_0) \mathbf{1} = \frac{x_1(0) + \dots + x_n(0)}{n} \mathbf{1}$$

$$= \bar{x}_0 \mathbf{1} \in \text{span}\{\mathbf{1}\}. \quad \blacksquare$$

Q What is the meaning of this theorem ?

① Centroid of motion is constant during convergence.

pf) Centroid $\bar{x}(t) = \frac{1}{n} (x_1(t) + \dots + x_n(t)) = \mathbf{1}^T x(t)$

$$\frac{d}{dt} \bar{x}(t) = \mathbf{1}^T \dot{x}(t) = -\mathbf{1}^T L(G) x(t)$$

Here, $L(G)\mathbf{1} = 0$ and $L(G)$ is symmetric,

$$L(G)^T \mathbf{1} = 0 \text{ and hence, } \mathbf{1}^T L(G) = 0$$

$$\therefore \frac{d}{dt} \bar{x}(t) = 0$$

Since we have init. condi. $x(0) = x_0$, $\bar{x}(t) = \frac{1}{n} \mathbf{1}^T x_0 \quad \forall t \geq 0$.

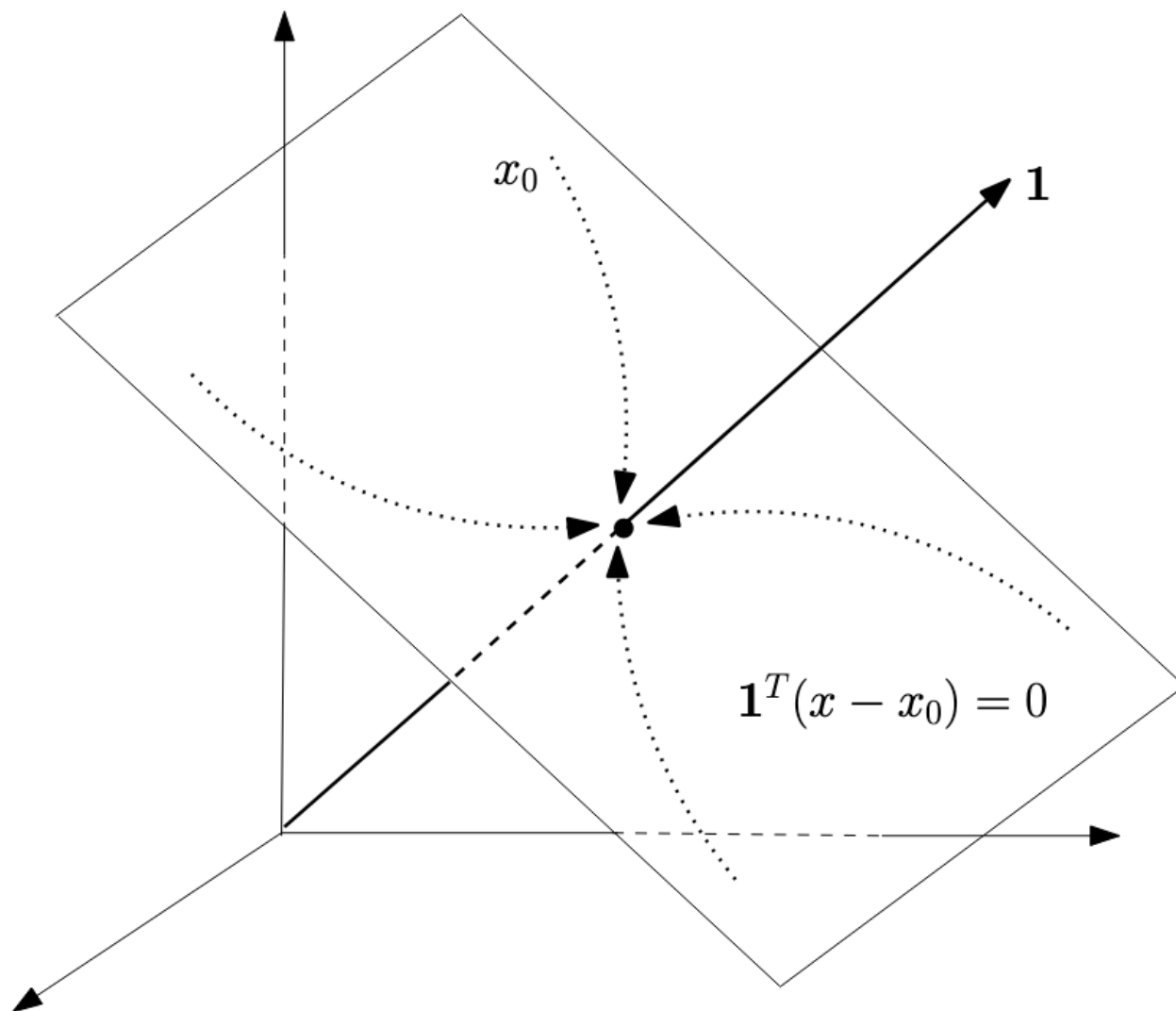
② The centroid is an **orthogonal projection** of init. state x_0 on the agreement space $\mathcal{A}$

Pf) Any vector $x \in \mathcal{A} = \text{span}\{\mathbf{1}\}$ can be written as $x = \alpha \mathbf{1}$.

$$\underset{\alpha}{\operatorname{argmin}} \|\alpha \mathbf{1} - x_0\| = \underset{\alpha}{\operatorname{argmin}} \|\alpha \mathbf{1} - x_0\|^2$$

$$\text{diff.} \rightarrow 2\mathbf{1}^T(\alpha \mathbf{1} - x_0) = 0 \Rightarrow \alpha = \frac{\mathbf{1}^T x_0}{\mathbf{1}^T \mathbf{1}} = \frac{1}{n} \mathbf{1}^T x_0$$

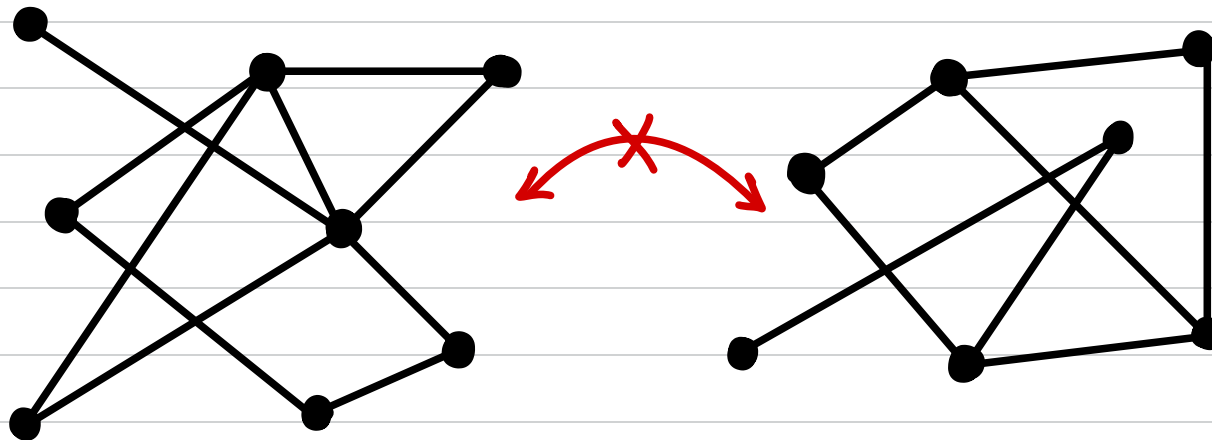
Thus, $x = \alpha \mathbf{1} = \frac{1}{n} \mathbf{1}^T x_0 \cdot \mathbf{1} \leftarrow$ Our convergence point!



Q. What is the **necessary and sufficient condition**
for a system to **reach consensus**?

(You may imagine the answer conceptually, w/o any mathematical logic)

What if the network topology is given by the following diagram?



No information Sharing!

Rigorously, from the eqn

$$x(t) = e^{-\lambda_1(G)t} u_1 u_1^T x_0 + e^{-\lambda_2(G)t} u_2 u_2^T x_0 \dots + \cancel{e^{-\lambda_n(G)t} u_n u_n^T x_0}, \quad \circ$$

Both $\lambda_1(G) = 0$, $\lambda_2(G) = 0$, the first two terms does not vanishing.

$$x(t) = u_1 u_1^T x_0 + u_2 u_2^T x_0$$

Since $u_1 \perp u_2$, and $u_1 \in \mathcal{A} = \text{span}\{1\}$,

$x(t)$ cannot converge to the agreement space.

Thm) Connected $\leftrightarrow$ Agree

A necessary and sufficient condition for the agreement protocol

$$\dot{x}(t) = -L(G)x(t)$$

Converge to the agreement space from an arbitrary init. condi.

is that G is connected.

= G contains a spanning tree.

Today's Paper :

Matrix - Weighted Consensus and its applications

Automatica (2018)

Minh Hoang Trinh, Chuong Van Nguyen,
Younghun Lim, Hyosung Ahn

- Now, we can move onto the **Matrix-weighted consensus** which **generalizes** scalar-weighted consensus.
- Introduce terminologies first. Most are just matrix analogue of the concepts that we already know.

Def) Matrix-weighted graph

The matrix-weighted graph is a **triple** $G = (V, E, A)$ where

$V = \{1, 2, \dots, n\}$ is a **vertex set** ;

$E = \{\{i, j\} : i, j \in V\}$ is an **edge set** ;

* Assume undirected, symmetric-weight

$A = \{A_{ij} \in \mathbb{R}^{d \times d} \cdot \{i, j\} \in E, A_{ij} = A_{ji} \succeq 0\}$ is **the set of weights**.

Def) Matrix-weighted Adjacency, Degree matrices and Laplacian.

Given a matrix-weighted graph $G = (V, E, A)$,

the matrix-weighted adjacency matrix is defined by $A = [A_{ij}]_{i,j \in [n]}$;

the matrix-weighted degree matrix is defined by $D = \text{bdiag}(D_i)$

where $D_i := \sum_{j \in N_i} A_{ij}$;

the matrix-weighted Laplacian is defined by $L = D - A$.

Def) Incidence matrix. (NOT new)

Given a matrix-weighted graph $G = (V, E, A)$ with $|V| = n$, $|E| = m$,

and order of edges $\sigma: [m] \rightarrow E$, the incidence matrix corresp. to

arbitrary orientation of E is defined by $H \in \mathbb{R}^{m \times n}$; $h_{ke} = \begin{cases} +1 & e=i \\ -1 & e=j \\ 0 & \text{othw} \end{cases}$

Def) Positive path, tree, spanning tree.

Given a matrix-weighted graph G .

- a positive path is a sequence of vertices in G denoted by

$$P = v_1 v_2 \dots v_n \text{ where } A_{v_i v_{i+1}} > 0 \quad \forall i \in [n-1].$$

Similarly, a positive tree/spanning tree is a tree/spanning tree

in G with PD edge weights.

Now, define a consensus protocol.

- Consider a system consisting of n single-integrator agents.
- i^{th} agent has a state vector $\mathbf{x}_i = [x_{i1}, \dots, x_{id}]^T \in \mathbb{R}^d$.
- The state vector of the system is $\mathbf{x} = [\mathbf{x}_1^T, \dots, \mathbf{x}_n^T]^T \in \mathbb{R}^{dn}$.

Def) Matrix-weighted Consensus protocol

$$\dot{\mathbf{x}}_i = \sum_{j \in N_i} A_{ij}(\mathbf{x}_j - \mathbf{x}_i) \text{ , i.e. } \dot{\mathbf{x}} = -\mathbf{L}\mathbf{x}.$$

The system is said to achieve consensus iff $\mathbf{x}_i = \mathbf{x}_j \ \forall i, j \in V$.

i.e. $\mathbf{x} \in \text{Range}\{\mathbf{1} \otimes \mathbf{I}_d\}$.

- There are two following results we need to prove.

Thm 1) Stability

Assume that G is positive-semiconnected. Then

any trajectory of the consensus protocol globally asymptotically approaches the set $N(\mathbb{I})$.

Thm 2) Average consensus (Algebraic condi. for reaching a consensus)

The system governed by the consensus protocol globally exponentially converges to the system's average

$x^* = \mathbb{1}_n \otimes \bar{x}$ if and only if $N(\mathbb{I}) = \mathbb{R}$.

Lem1) Decomposition of matrix-weighted Laplacian

$$\mathbb{L} = \bar{H}^T \text{bdiag}(A_k) \bar{H}$$

Lem2) Properties of matrix-weighted Laplacian

$\mathbb{L}$ is (i) Symmetric (ii) PSD.

$$(iii) N(\mathbb{L}) = \{v \in \mathbb{R}^{dn} \cdot (v_j - v_i) \in N(A_{ij}), \forall \{i,j\} \in \mathcal{E}\}$$

Lem3) Invariance of the average

Under the matrix-weighted consensus protocol $\dot{\mathbf{x}} = -\mathbf{L}\mathbf{x}$,

the average $\bar{\mathbf{x}} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i$ is invariant.

Proof) Note that $\bar{\mathbf{x}} = \frac{1}{n} (\mathbf{1}_n^T \otimes \mathbf{I}_d) \mathbf{x}$.

$$\begin{aligned} \text{Differentiate: } \dot{\bar{\mathbf{x}}} &= \frac{1}{n} (\mathbf{1}_n^T \otimes \mathbf{I}_d) \dot{\mathbf{x}} \\ &= \frac{1}{n} (\mathbf{1}_n^T \otimes \mathbf{I}_d) (-\mathbf{L}\mathbf{x}) = -\frac{1}{n} (\mathbf{1}_n^T \otimes \mathbf{I}_d) \mathbf{L}\mathbf{x} \end{aligned}$$

By lemma 1. $N(\mathbf{L}) \supseteq N(\bar{\mathbf{H}})$, and obviously, $\bar{\mathbf{H}}(\mathbf{1}_n \otimes \mathbf{I}_d) = \mathbf{0}$ and hence

$\mathbf{R} \not\subseteq N(\bar{\mathbf{H}})$. Thus, $\mathbf{R} \not\subseteq N(\mathbf{L})$.

So, $\mathbf{0} = \mathbf{L}(\mathbf{1}_n \otimes \mathbf{I}_d) = \mathbf{L}(\mathbf{1}_n \otimes \mathbf{I}_d^T)$.

Take transpose. $\mathbf{0} = (\mathbf{1}_n \otimes \mathbf{I}_d^T)^T \mathbf{L}^T = (\mathbf{1}_n^T \otimes \mathbf{I}_d) \mathbf{L}$.

$$\therefore \dot{\bar{\mathbf{x}}} = \mathbf{0}. \quad \blacksquare$$

Proof of thm 1)

Consider a PSD, radially unbounded C^1 function $V(x) = \frac{1}{2} \|x\|^2$.

$$V = \frac{1}{2} x^T x \text{ and hence } \dot{V} = x^T \dot{x} = x^T (-Lx) = -x^T Lx.$$

Since $L \succeq 0$, we have $\dot{V} \leq 0$.

Thus, $\|x(t)\| \leq \|x(0)\|$ i.e. x is bounded.

x bounded, $\dot{V} \leq 0 \Rightarrow \dot{V}(x(t)) \rightarrow 0$ as $t \rightarrow \infty$

by LaSalle's invariance principle.

Since $\dot{V} = 0 \iff x \in N(L)$, equivalently, we can say that

$x(t) \rightarrow N(L)$ (i.e. $x(\infty) \in N(L)$) asymptotically.

Since the initial point can be freely chosen, such convergence is global.

Lem4) Uniqueness of Equilibrium

If $N(L) = R$, the system governed by matrix-weighted consensus protocol has the only equilibrium point $x^* = \mathbf{1}_n \otimes \bar{x}$.

Sketch of proof) Suppose NOT. $\exists$ another equilibrium $x' \neq x^*$

Note that $\bar{x}' = \frac{1}{n}(\mathbf{1}_n^T \otimes I_d)x'$ and by lemma 3, $\bar{x}' = \bar{x}$.

Thus, $x^* = x'$. $\Rightarrow \Leftarrow$.

Proof of thm2)

($\Rightarrow$) Suppose NOT. $x(t)$ globally asymptotically converges to x^*
but NOT $N(L) = \mathbb{R}$.

From the proof of lemma 3, $\exists x' \in \mathbb{R}^{d_n}$ s.t. $x' \in N(L) \setminus \mathbb{R}$.

Then, $Lx' = 0$ and hence $\dot{x}' = -x'^T Lx' = 0$.

Thus, x' be another equilibrium point and hence contradicts that equilibrium point x^* is NOT globally asymptotically stable. $\rightarrow \leftarrow$.

($\Leftarrow$) Suppose $N(\mathbb{L}) = \mathcal{R}$.

By lemma 4, $x^* = \mathbf{1}_n \otimes \bar{x} \in N(\mathbb{L})$ is the unique equilibrium point.

Consider the Lyapunov function

$$V = \frac{1}{2} \|\delta\|^2 \text{ where } \delta = x - \mathbf{1}_n \otimes \bar{x} \text{ (disagreement measure)}$$

Then, V is PSD, radially unbounded, C^1

$$\begin{aligned} \text{and } \dot{V} &= \delta^T \dot{\delta} = -\delta^T \mathbb{L} x = -\delta^T \mathbb{L} (\delta + \mathbf{1}_n \otimes \bar{x}) \\ &= -\delta^T \mathbb{L} \delta - \underbrace{\delta^T \mathbb{L} (\mathbf{1}_n \otimes \bar{x})}_{=0} = -\delta^T \mathbb{L} \delta \end{aligned}$$

Since $\mathbb{L}$ is PSD, $\dot{V} \leq 0$.

Moreover, since δ measures disagreement, $\underline{\delta \perp \mathcal{R}}$ and hence, ①

$$\dot{V} = -\delta^T \mathbb{L} \delta \stackrel{\textcircled{2}}{\leq} -\lambda_{d+1}(\mathbb{L}) \delta^T \delta \leq -\alpha V \text{ (for some } \alpha)$$



Details

① $\delta \perp \mathbb{R}$

$$\begin{aligned} \langle (\mathbb{1}_n \otimes I_d), \delta \rangle &= (\mathbb{1}_n \otimes I_d)^T \delta \\ &= (\mathbb{1}_n \otimes I_d)^T \mathbb{X} - (\mathbb{1}_n^T \mathbb{1}_n \otimes I_d) \bar{\mathbb{X}} \\ &= n\bar{\mathbb{X}} - n\bar{\mathbb{X}} = 0. \end{aligned}$$

② Why $-\delta^T \mathbb{L} \delta \leq -\lambda_{d+1}(\mathbb{L}) \delta^T \delta$?

Rayleigh quotient property: $\lambda_{d+1}(\mathbb{L}) = \min_{\delta \perp \mathbb{R}} \frac{\delta^T \mathbb{L} \delta}{\delta^T \delta} \leq \frac{\delta^T \mathbb{L} \delta}{\delta^T \delta}$

Recall that **theorem 2** algebraically characterizes the condi.
for reaching a consensus.

Q. Is there any **combinatorial** condi.?

Previously, in scalar consensus protocol, we have

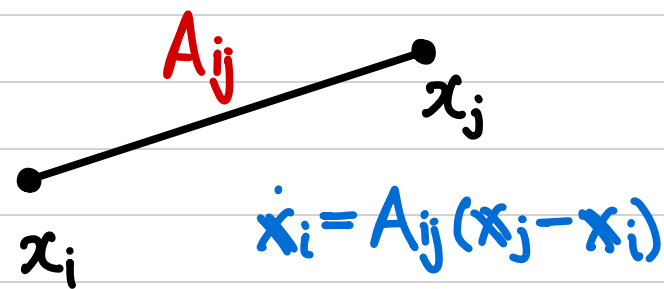
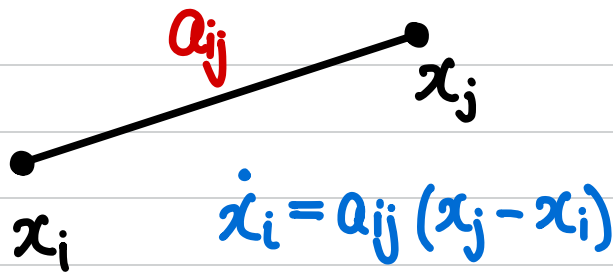
G is connected $\iff x(t) \rightarrow \bar{x}$ as $t \rightarrow \infty$ under $\dot{x} = -L(G)x$

↳ Can it be generalized as well?

Lem 5) Combinatorial Sufficient condi. for reaching a consensus.

If G has a **positive spanning tree** $\mathcal{T}$, the n -agent system **globally exponentially achieves an average consensus** under the consensus protocol

↳ Why sufficient condi. only?



Example) $x_i = [0 \ 0]^T$, $x_j = [0 \ 5]^T$ and $A_{ij} = \text{diag}(1, 0)$.

Def) Clusters

Given a positive tree T , the cluster $C(T)$ generated by T

is defined by $C(T) := V(T) \cup X(T)$ where

$$X(T) = \{i \in V(G) \setminus V(T) : \text{condi. (a). (b)}\}.$$

To state condi. (a) (b), define a set S_i first.

$$S_i := \{P_k = \{v_1^k, \dots, v_{|P_k|}^k\} : v_1^k = i, v_{|P_k|}^k \in V(T), \forall j = 1, \dots, |P_k| - 1, v_j^k \notin V(T)\}$$

i.e. the set of walks from i reaching T .

Now, conditions to belong $X(T)$ be:

(a) $\forall P_k \in S_i$, $\dim\left(\bigcap_{k=1}^{|S_i|} N(P_k)\right) = 0$ where $N(P_k) := \sum_{j=1}^{|P_k|-1} N(A_{v_j^k v_{j+1}^k})$

(b) P_k is a path, i.e. $\nexists$ cycle.

Thm3) Cluster consensus.

Under the matrix-weighted consensus protocol, **all agents in the cluster $C(T)$ have the same equilibrium state.**

In other words, given an equilibrium state x^* , $x_i^* = x_j^* \quad \forall i, j \in C(T)$.

Thm4) Average consensus

If G is spanned by a cluster, then **average consensus is reached** under the matrix-weighted consensus protocol.

* Note: G is spanned by $C(T)$ if $V(T) = V(G)$.

Proof of thm4)

If G is spanned by a cluster, by thm3, all agents share an equilibrium state $x^* = \mathbf{1}_n \otimes \bar{x}$.

Since $x^* \in N(\mathbb{L})$, this implies that $N(\mathbb{L}) = \mathbb{R}$ and hence, by the thm 2, x reaches an average consensus. $\blacksquare$

Thank You

It's the end of the presentation.